

SAGAR MEHTA

Technical Product Lead | Product Development & Hardware Systems

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Product development and systems engineering leader with 8+ years of experience bringing complex hardware products from concept through commercialization, certification, manufacturing, and deployment. Proven track record leading cross-functional teams across hardware, software, firmware, quality, regulatory, and manufacturing disciplines. Experienced translating customer and business requirements into technical roadmaps, product requirements, validation plans, and market-ready solutions. Strong background in product strategy, stakeholder management, and commercialization of innovative technology platforms.



Skills

Systems Integration, Hardware Verification & Validation (V&V), Product lifecycle management, Regulatory & certification planning, Root Cause Analysis (RCA), Sensor Integration, Linux Based Embedded Systems, Mechanical Design, Rapid Prototyping, DFMEA, DFM, Regulatory Compliance, Hardware-Software Integration, Technical product strategy, Reliability Testing, Cross-functional Leadership

Tools: Python, C, MATLAB, Solidworks, Simulink, 3D Printing & Prototyping, Git, Embedded Linux, Communication Protocols, Quality Management System, Claude Code, MS office tools



Work Experience

2020-06 -
Current

Lead Product Development Engineer

Cannabix Technologies Inc., Burnaby

- Led development of alcohol and THC detection platforms from concept through regulatory approval, manufacturing readiness, and commercial deployment.
- Collaborated with executive leadership, engineering, manufacturing, and regulatory teams to prioritize product features, development milestones, and market launch objectives.
- Translated customer, regulatory, and operational requirements into system specifications and product development roadmaps.

- Managed technical trade-offs between performance, manufacturability, reliability, cost, and project timelines.
- Supported product commercialization activities including certification planning, manufacturing transfer, supplier engagement, and field deployment.
- Supported certification activities and regulatory validation for commercial breath-testing products deployed in safety-critical environments.

2018-12 -
2020-05

Mechatronics Engineer

CARPA 3D Construction, Burnaby

- Designed and integrated actuators, sensors, motor drives, embedded controllers, and control systems for a large-scale robotic concrete printing platform.
- Developed and executed hardware validation testing to characterize motion accuracy, repeatability, actuator performance, and structural behavior.
- Integrated encoder systems, limit switches, motor controllers, and safety interlocks to support reliable autonomous operation.
- Performed system-level debugging across mechanical, electrical, and software domains to identify and resolve integration issues.
- Designed and fabricated custom brackets, fixtures, cable management systems, and mechanical assemblies using rapid prototyping techniques.
- Built and validated a full-scale robotic construction platform capable of printing structures with ± 5 mm positional accuracy.
- Contributed directly to development of the company's first operational prototype which helped secure \$2M in investor funding.

2015-09 -
2018-04

Research Assistant

Simon Fraser University, Burnaby

- Developed and validated control algorithms using MATLAB and Simulink for dynamic mechanical systems.
- Designed experimental setups, collected data, and analyzed system performance to validate modeling assumptions.
- Published peer-reviewed research focused on control systems and system stability in 4 highly distinct journals.

2015-01 -
2015-05

Research and Development Intern

Bosch-Rexroth, Pune

- Organized and handled project requirements, resource identification, and task execution.
- Corroborated the design, efficiency and performance of the unit with the team at Bosch Rexroth
- Successfully developed a modular unit with higher energy savings that can be used in actuation of small cylinders, fixture clamps, and other machine tool applications
- An overall energy saving of up to 82 percent was achieved covering over conventional units



Education

2018-04 **Master of Applied Science: Mechatronic Systems Engineering**

Simon Fraser University - Burnaby, BC

2015-05 **Bachelor of Engineering: Mechanical Engineering**

PES Institute of Technology - Bengaluru, KA



Accomplishments

- Owned end-to-end technical contributions for **patented** and **certified** alcohol and THC breath-testing devices, directly supporting **regulatory approval** and **market launch**.
- Spearheaded the design, prototyping, and validation of a full-scale concrete 3D printer, a functional robotics system that secured **\$2M in investor funding** and accelerated company growth into large-scale construction automation.
- Published **four** peer-reviewed research papers in reputable scientific **journals**, showcasing expertise in control systems engineering and innovation.



Publications

- **Adaptive sliding mode observers for sector-bounded nonlinear systems**

(International Journal of Adaptive Control and Signal Processing
Aug 21, 2021)

- **Design of Sliding Observers for Lipschitz Nonlinear System Using a New Time-averaged Lyapunov Function**

(International Journal of Control · May 21, 2018)

- **Analysis of Sliding Mode Observers for Nonlinear Systems Using a Time-averaged Lyapunov Function**

(ASME 2018 International Design Engineering Technical Conferences
and Computers and Information in Engineering Conference · Aug 1,
2018)

- **Analysis of Sliding Mode Observers Using a Novel Time-Averaged Lyapunov Function**

(ASME 2017 International Design Engineering Technical Conferences
and Computers